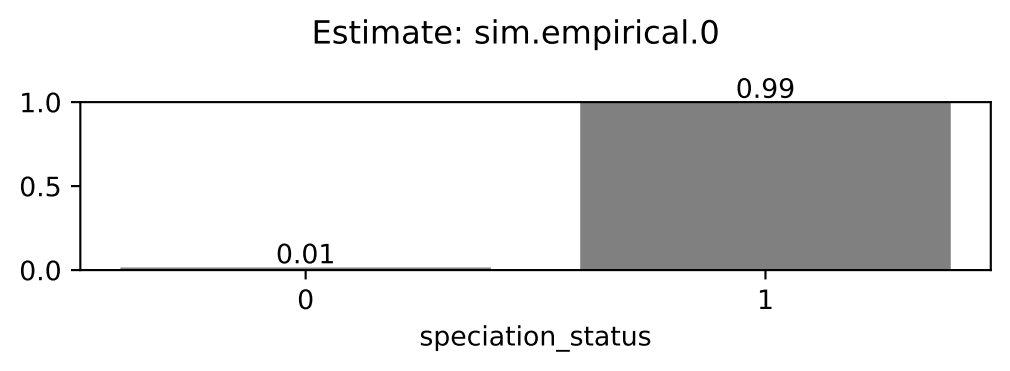
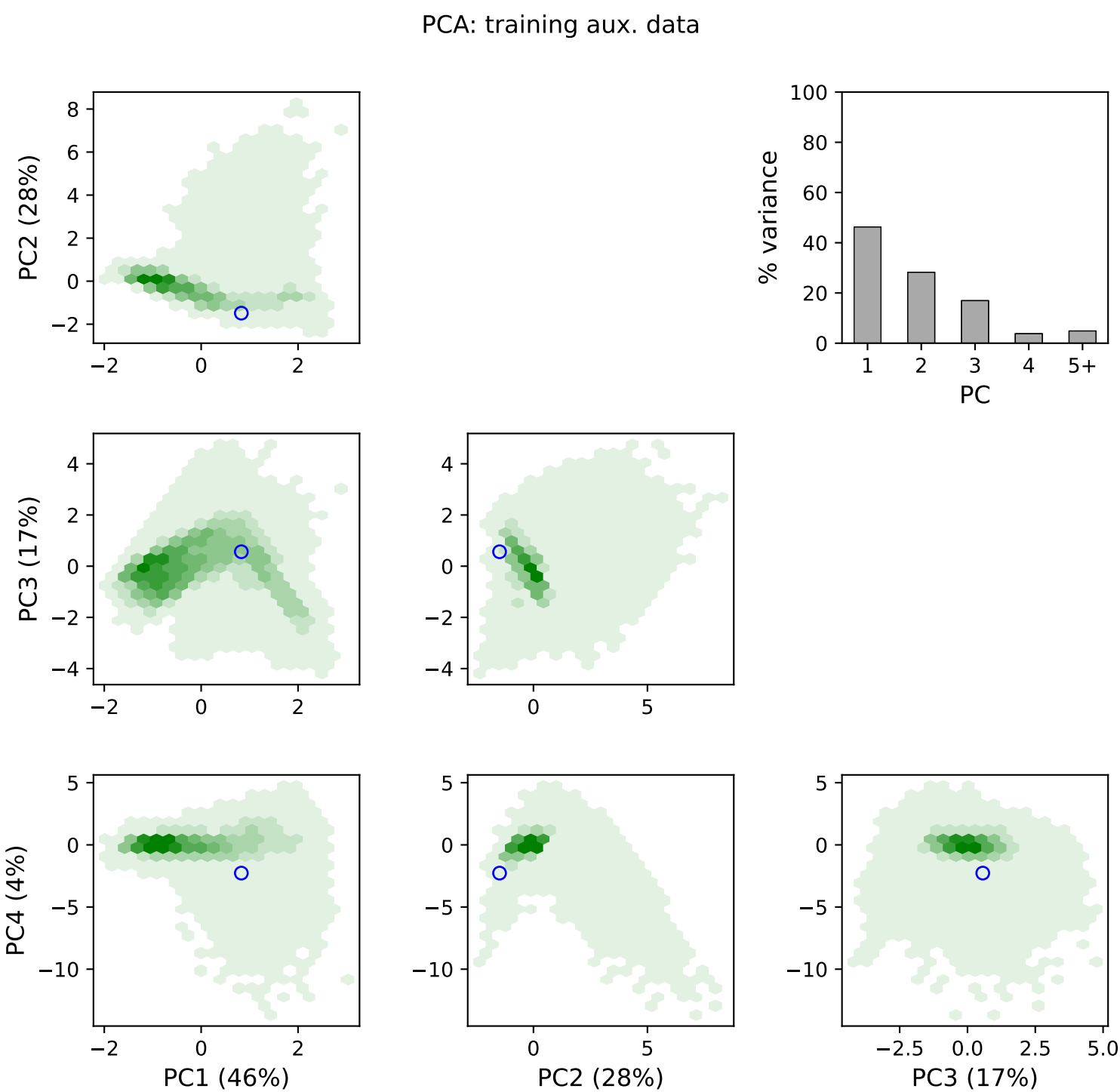
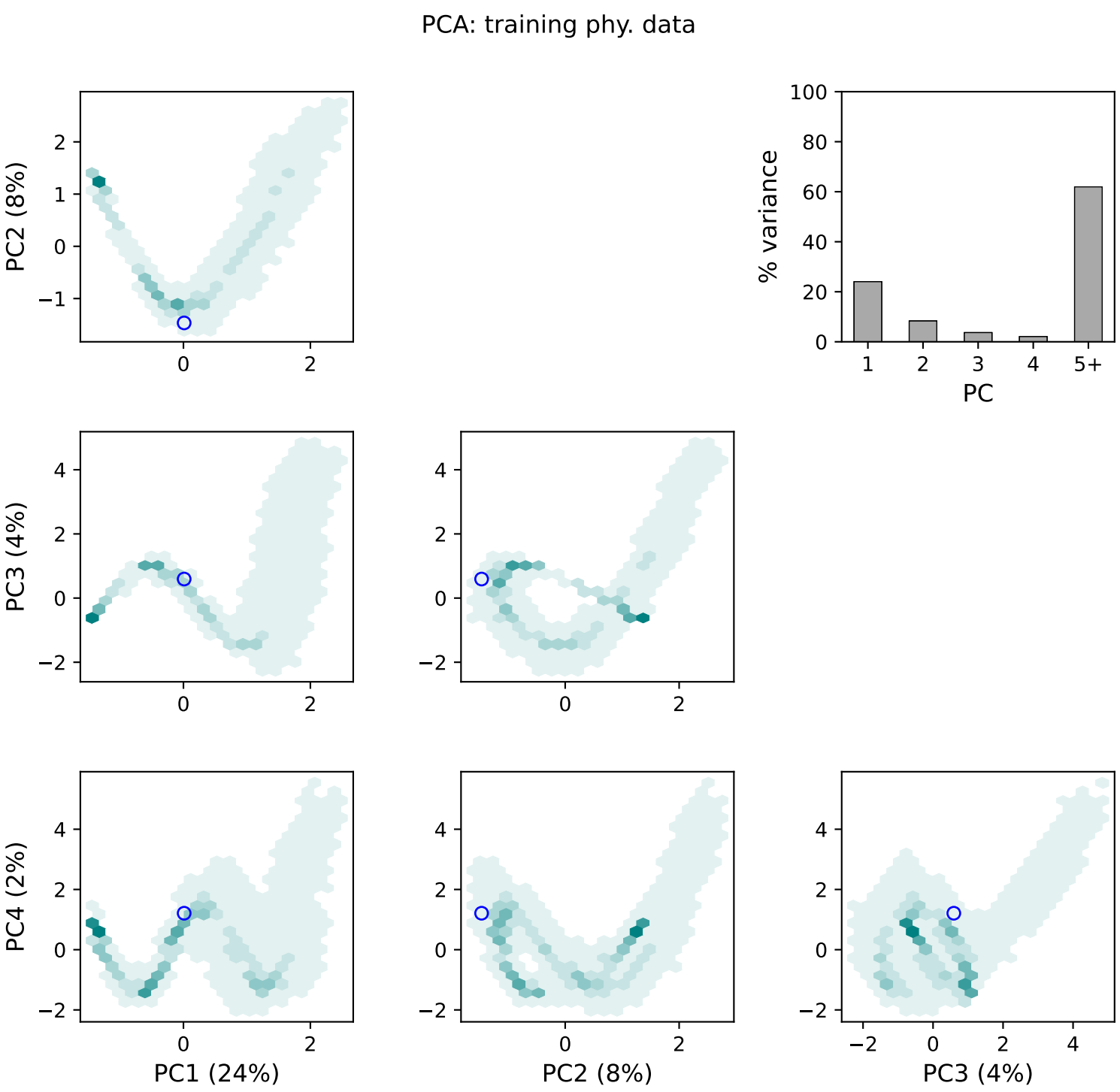
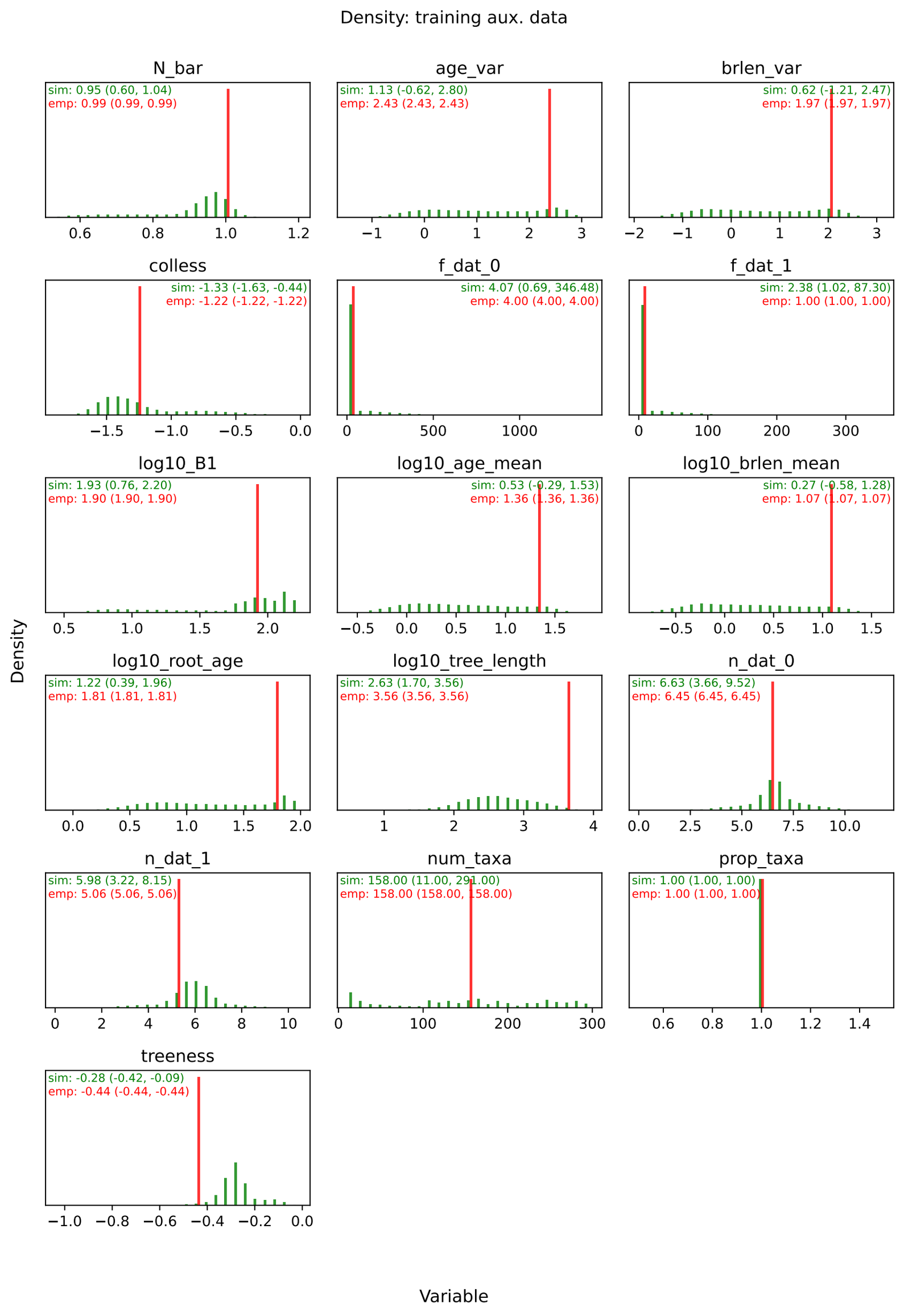
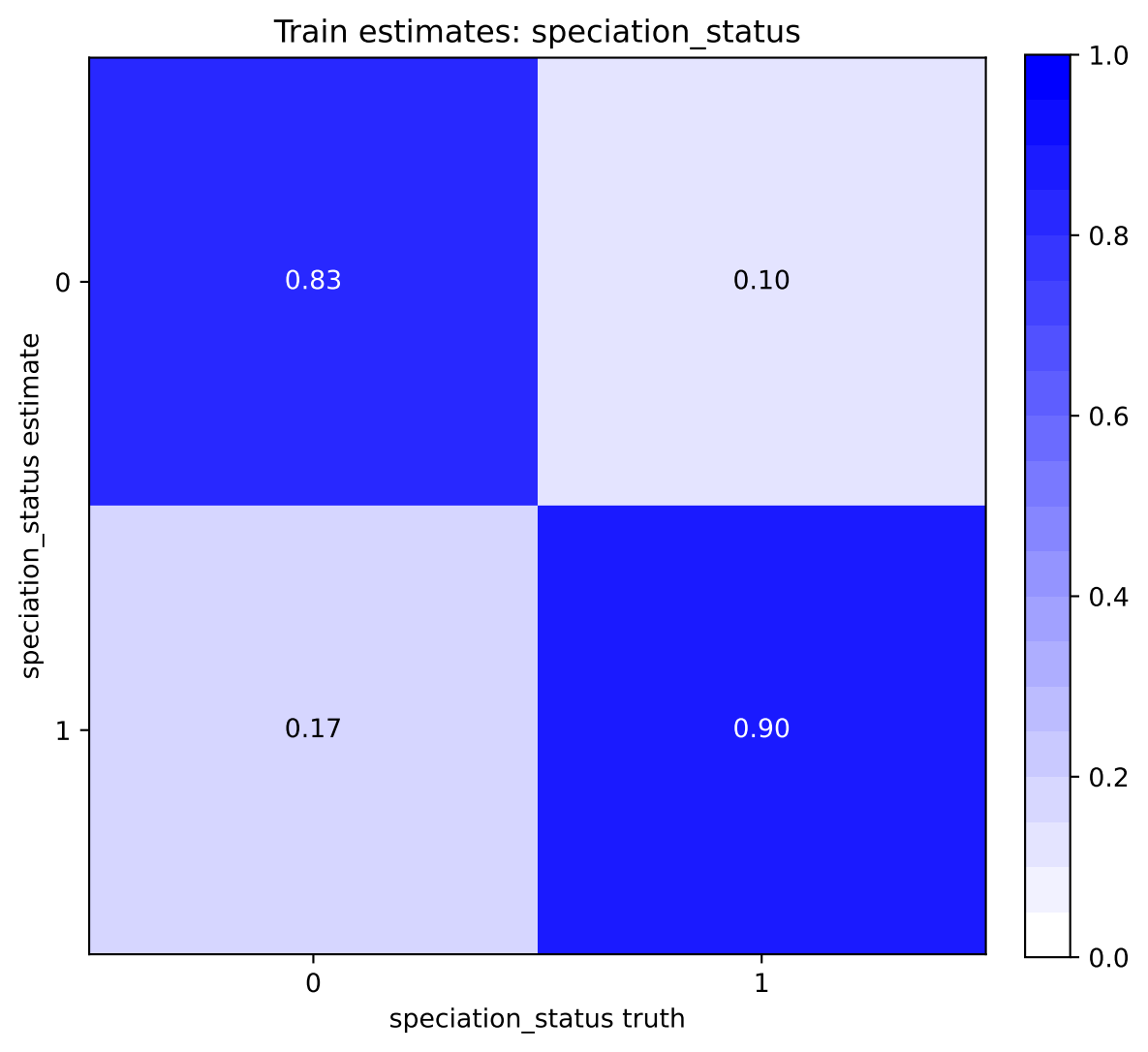
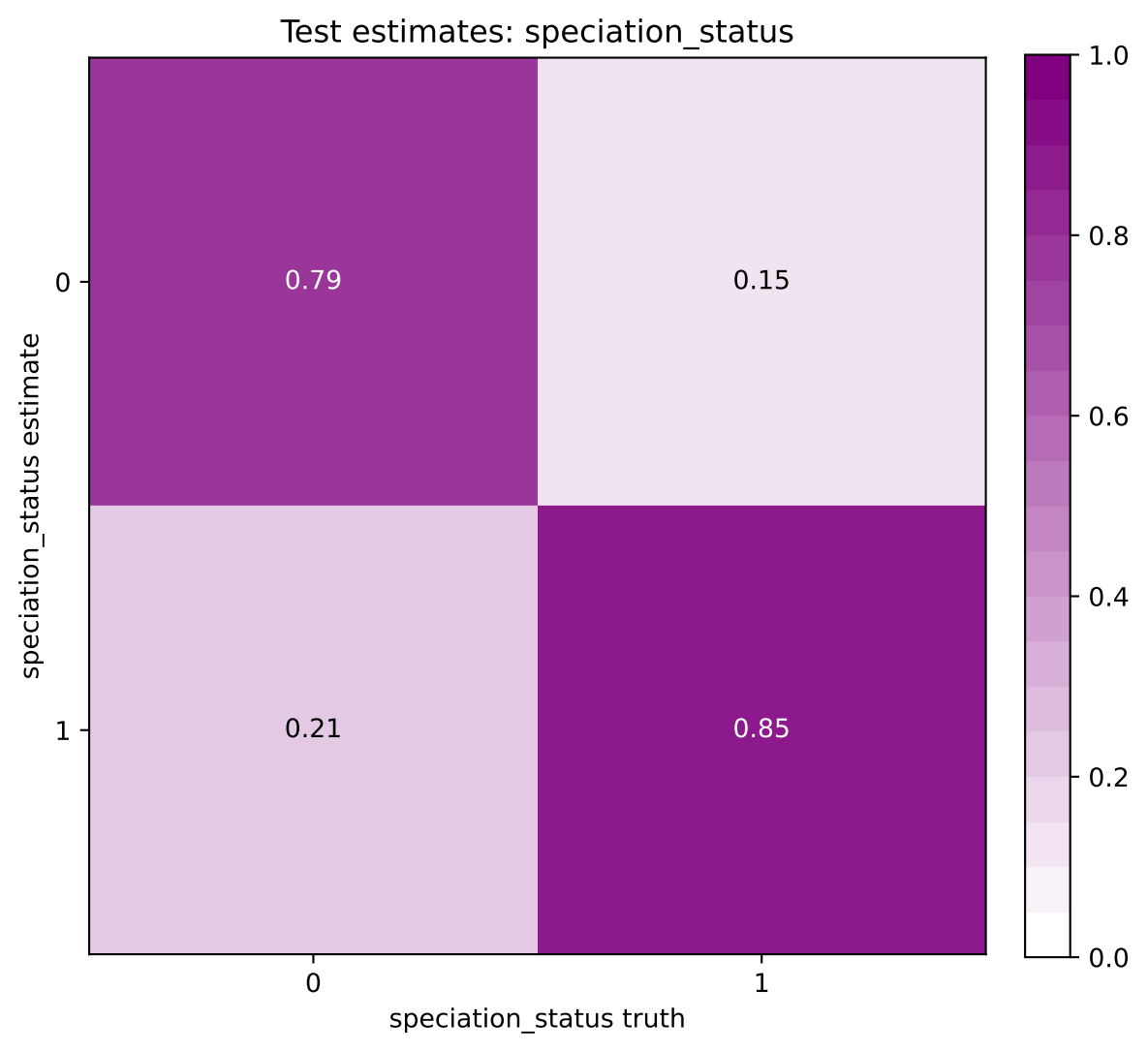

Estimate: sim.empirical.0
1.0
0.5
0.0
0.99
0.01
0
1
speciation_status

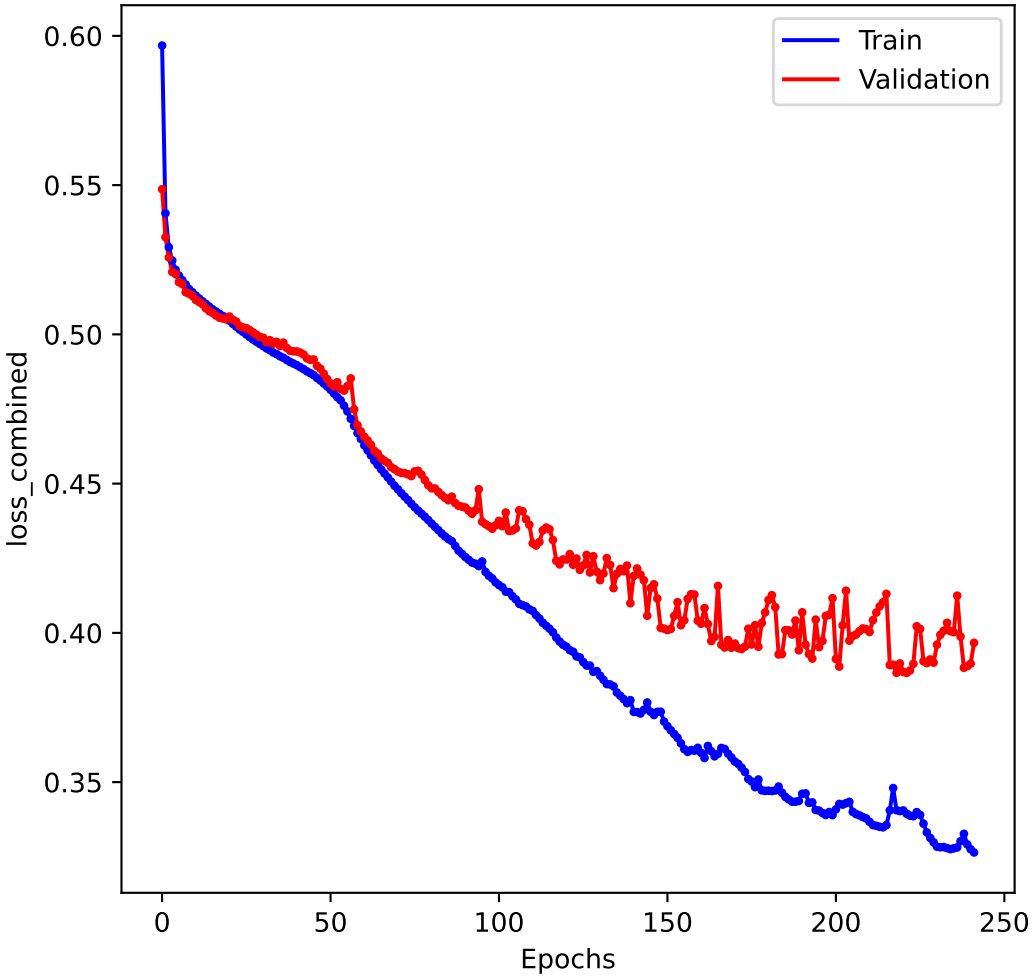

PCA: training aux. data
PC2 (28%)
% variance
PC
PC3 (17%)
PC4 (4%)
PC1 (46%)
PC2 (28%)
PC3 (17%)

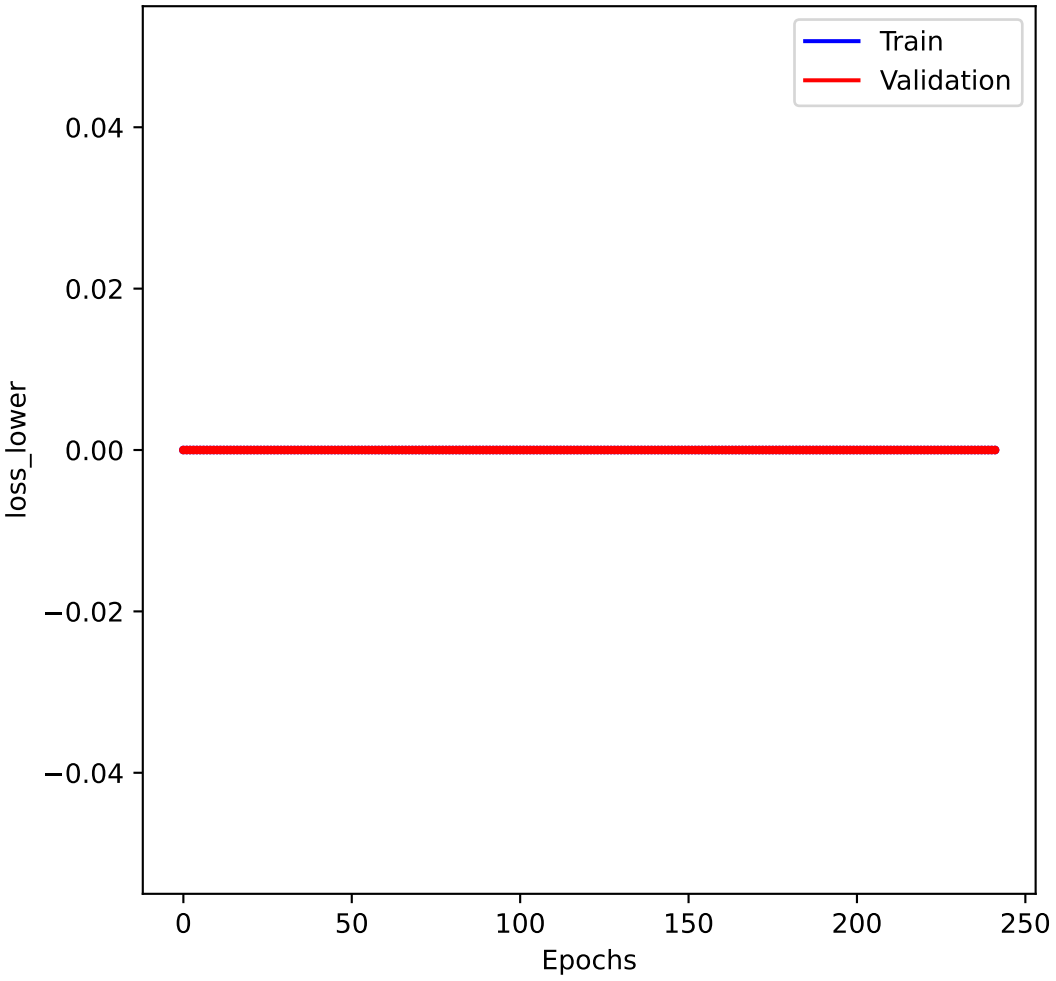

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (24%)
PC2 (8%)
PC3 (4%)

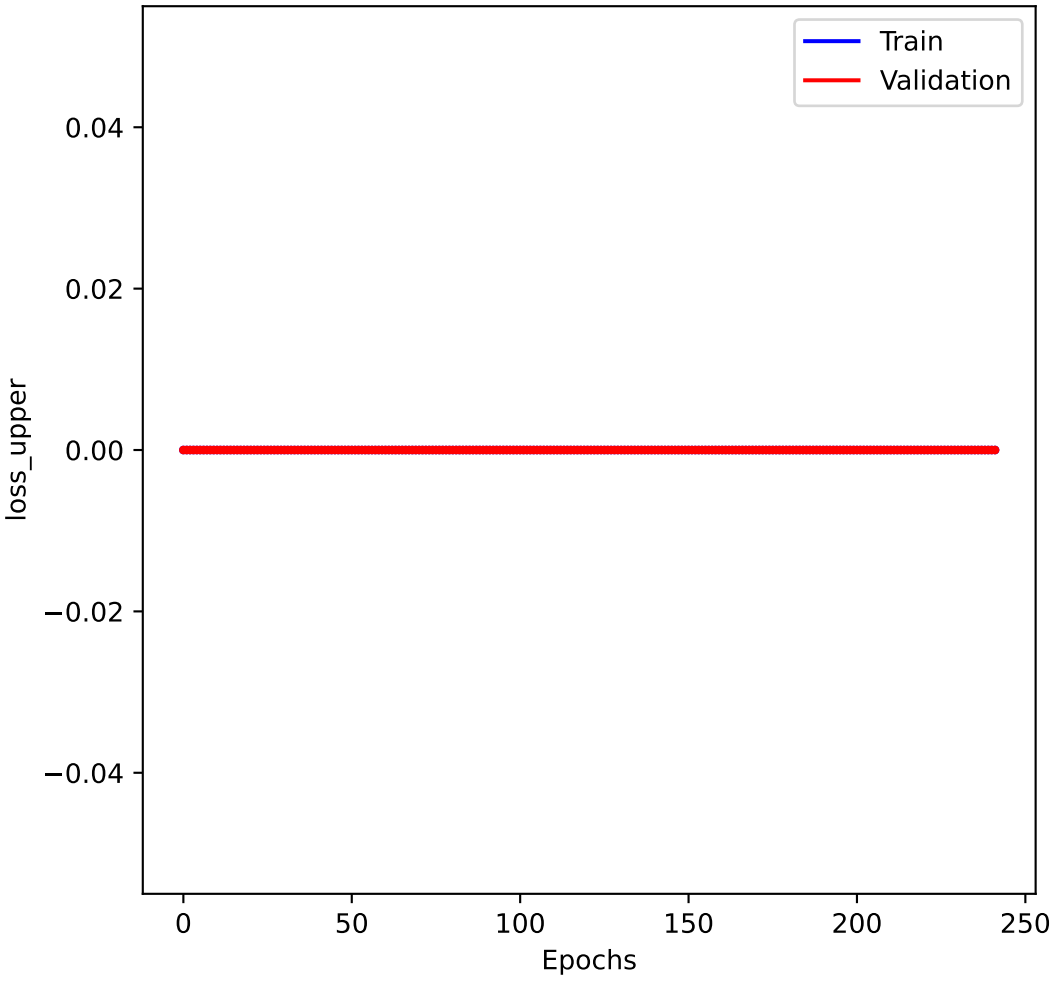

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.13 (-0.62, 2.80)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.62 (-1.21, 2.47)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.33 (-1.63, -0.44)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.07 (0.69, 346.48)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.38 (1.02, 87.30)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.93 (0.76, 2.20)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.53 (-0.29, 1.53)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.27 (-0.58, 1.28)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.22 (0.39, 1.96)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.63 (1.70, 3.56)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.63 (3.66, 9.52)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.98 (3.22, 8.15)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 158.00 (11.00, 291.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.42, -0.09)
emp: -0.44 (-0.44, -0.44)
Density
Variable

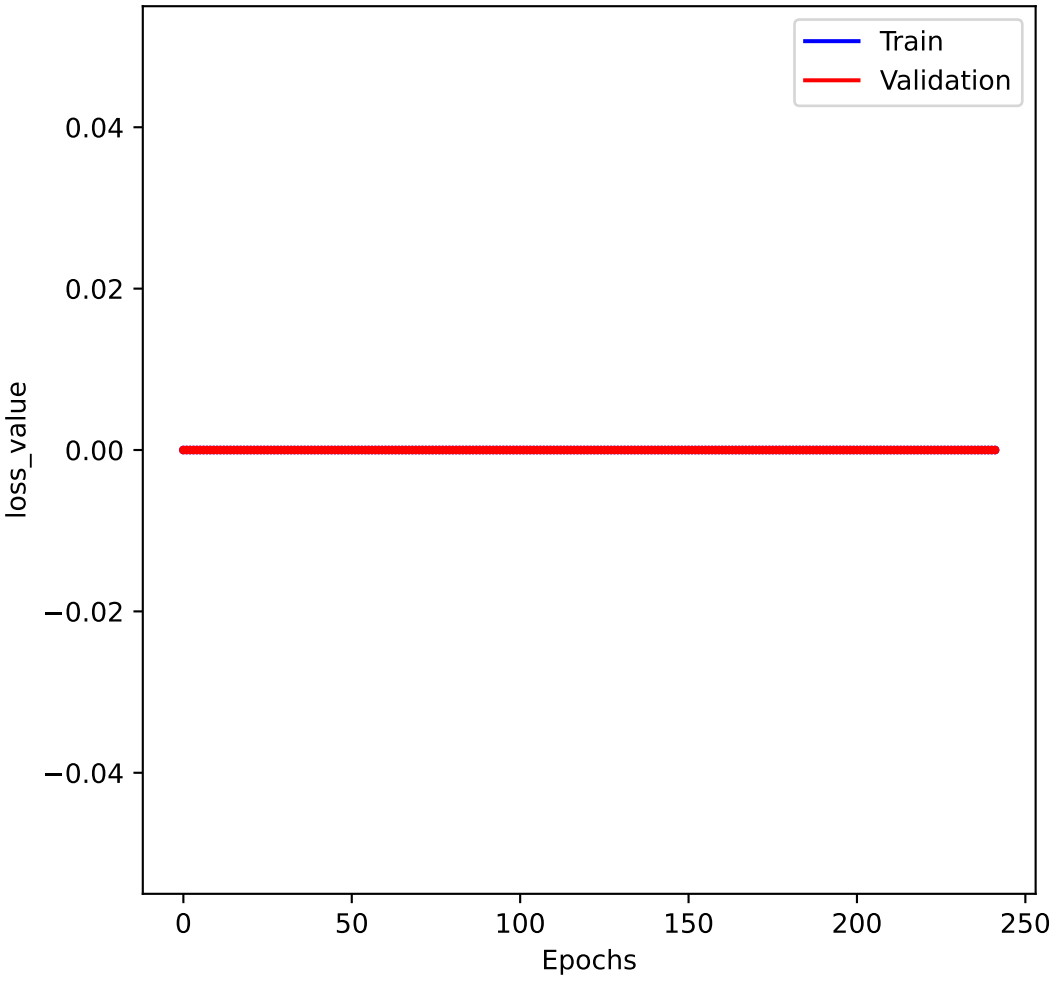

Train estimates: speciation_status
0.83
0.10
0.17
0.90
0
1
speciation_status estimate
0
1
speciation_status truth
1.0
0.8
0.6
0.4
0.2
0.0

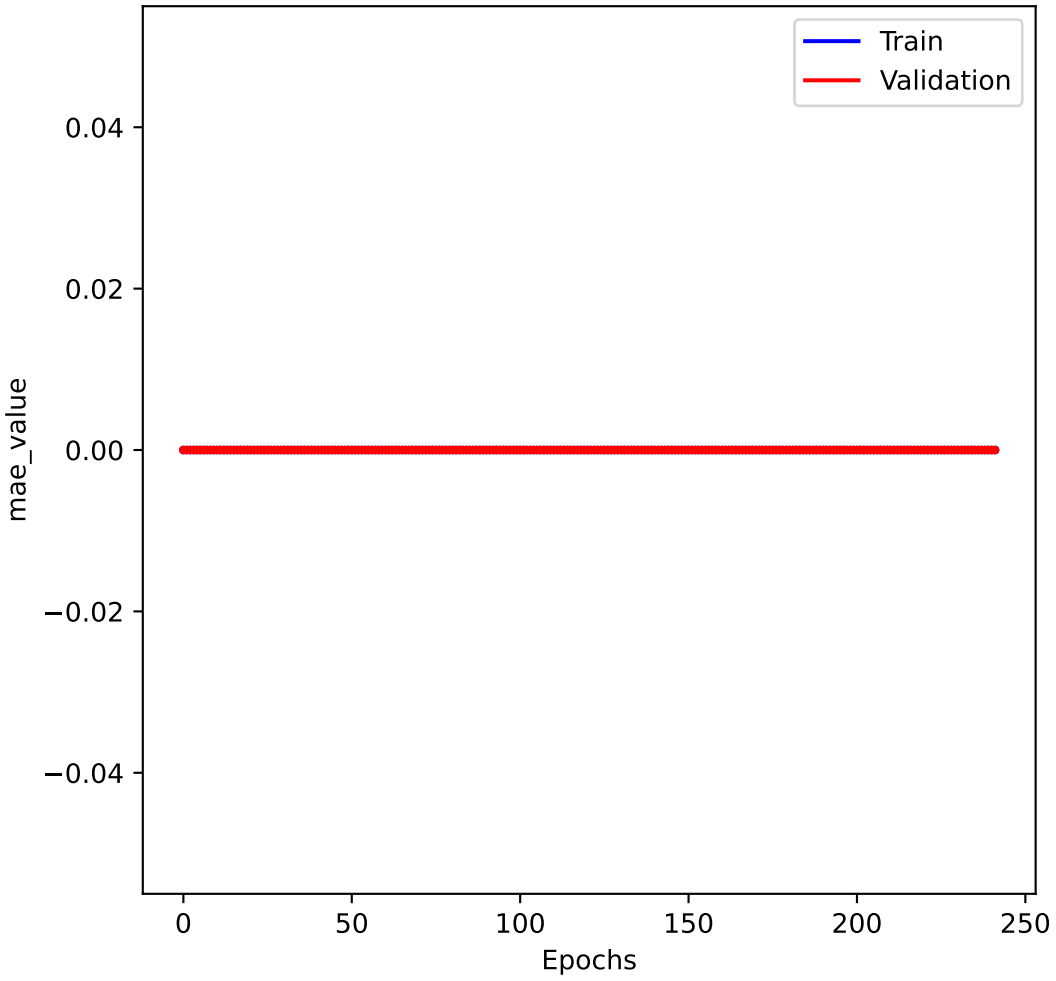

Test estimates: speciation_status
0.79
0.15
0.21
0.85
0
1
speciation_status estimate
0
1
speciation_status truth
1.0
0.8
0.6
0.4
0.2
0.0

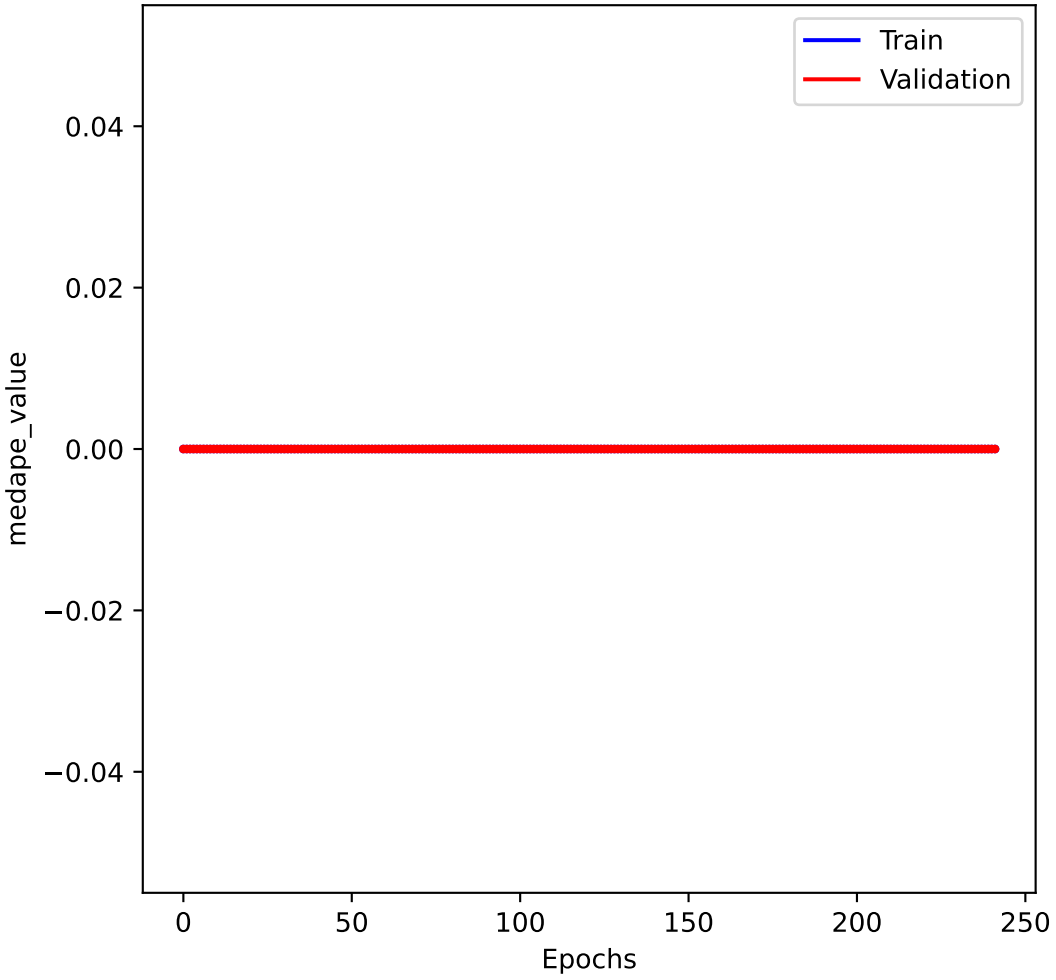

Train
Validation
loss_combined
0.60
0.55
0.50
0.45
0.40
0.35
0
50
100
150
200
250
Epochs

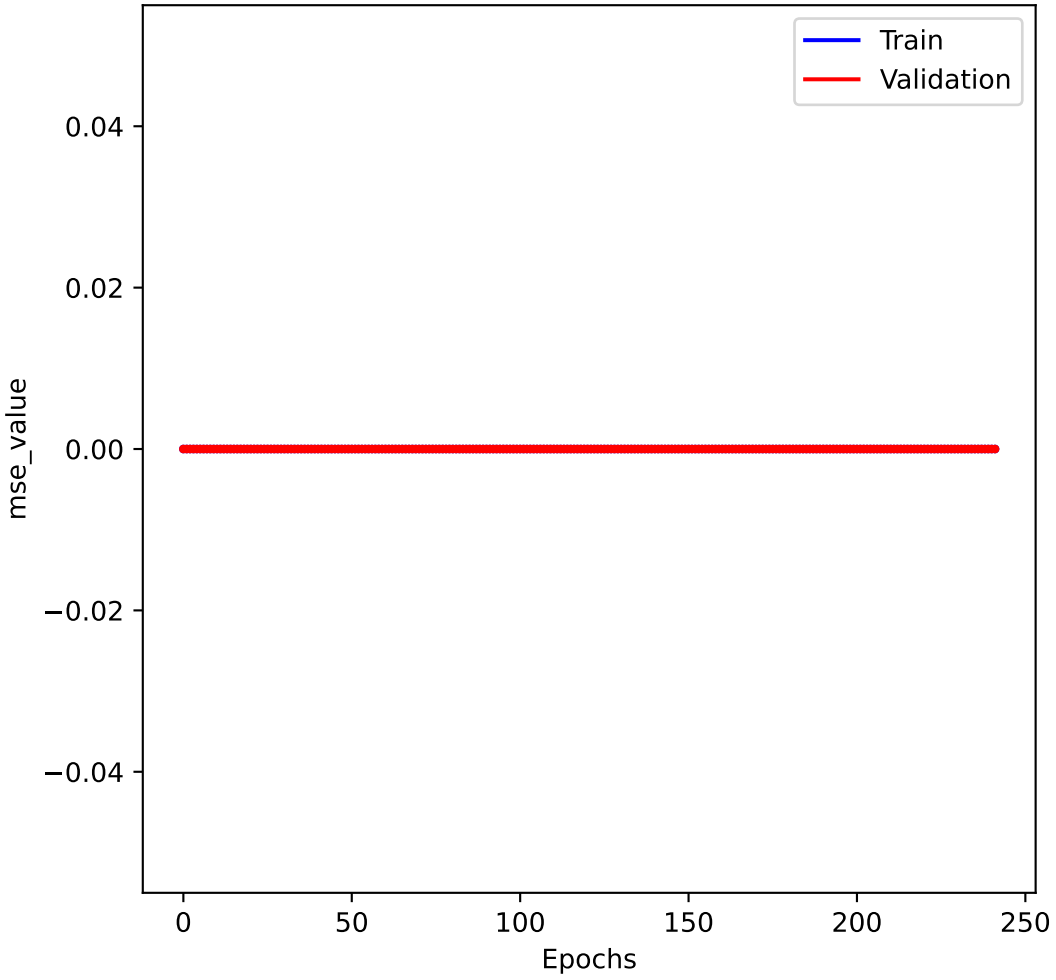

Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_lower
0
50
100
150
200
250
Epochs

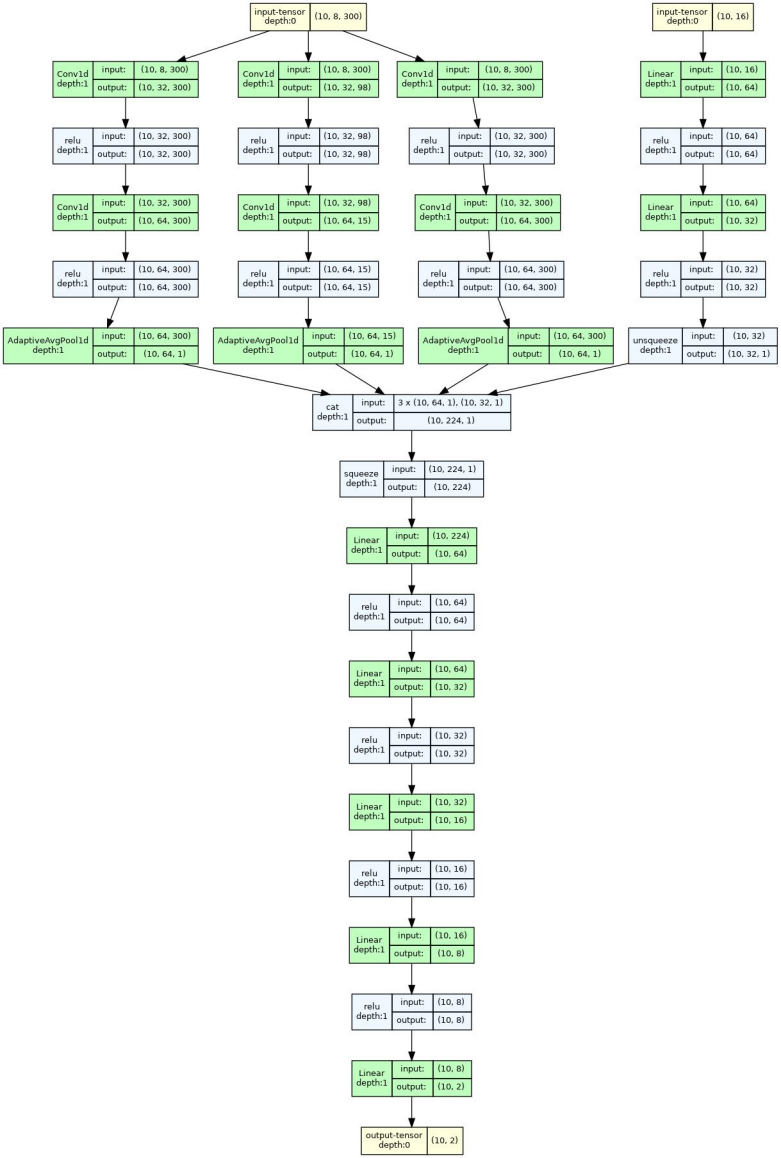

Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_upper
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
mae_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
medape_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
mse_value
0
50
100
150
200
250
Epochs


input-tensor depth:0 (10, 8, 300)
input-tensor depth:0 (10, 16)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 98)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Linear depth:1 input: (10, 16) output: (10, 64)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 32, 98) output: (10, 32, 98)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Conv1d depth:1 input: (10, 32, 98) output: (10, 64, 15)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64, 15) output: (10, 64, 15)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 15) output: (10, 64, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: 3 x (10, 64, 1), (10, 32, 1) output: (10, 224, 1)
squeeze depth:1 input: (10, 224, 1) output: (10, 224)
Linear depth:1 input: (10, 224) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 16)
relu depth:1 input: (10, 16) output: (10, 16)
Linear depth:1 input: (10, 16) output: (10, 8)
relu depth:1 input: (10, 8) output: (10, 8)
Linear depth:1 input: (10, 8) output: (10, 2)
output-tensor depth:0 (10, 2)